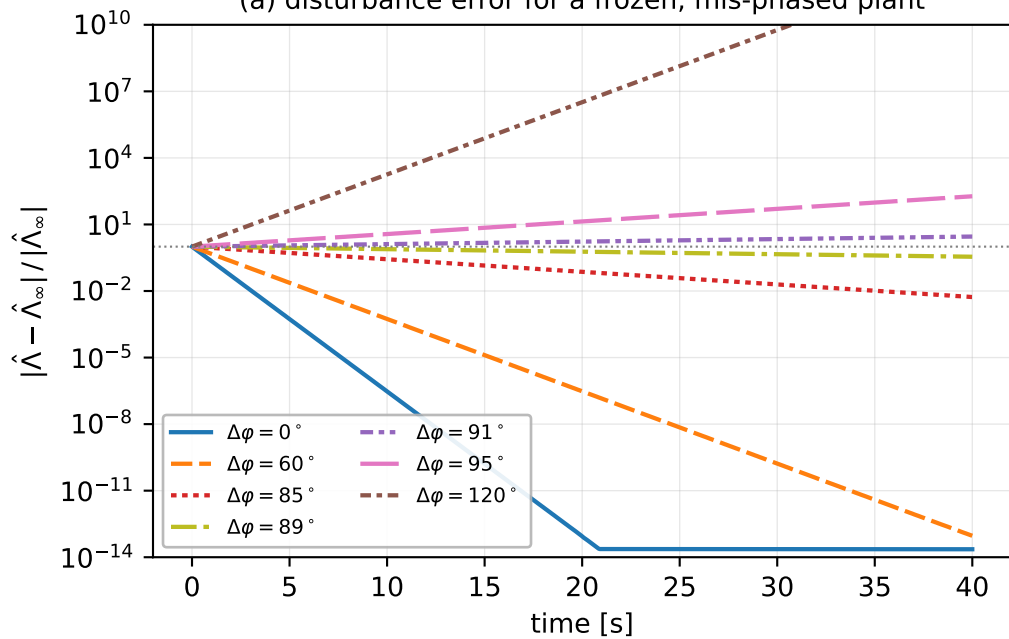


(a) disturbance error for a frozen, mis-phased plant

(b) rate changes sign exactly at  $90^\circ$ 